Week 06 — Networking Fundamentals for Security

> Where data travels and where security matters. Slides: week06-networking-fundamentals-for-security.pdf

By the end you should be able to:

1. Name the devices and follow a network path.
2. Explain IP addresses, ports, protocols, and DNS.
3. Locate a risk and choose a suitable control.

Opening question

> A student opens the university LMS on café Wi-Fi. The browser shows a **certificate warning**. *Where could the problem be?* Discuss the **device**, the **network**, and the **website**.

The point of the question is that a warning doesn't tell you *where* the problem is. It could be:

- **the device** (wrong clock, malware, or software intercepting traffic),
- **the network** (a fake hotspot or a portal intercepting the connection),
- **the website** (an expired or wrong certificate on the real server).

The habit the whole lecture builds: **name the location first**, then the weakness, then the control.

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1. Follow the network path

> A **network** connects devices so they can exchange data.

One visit to the LMS

The laptop **sends a request**, the server **sends a response**. In the simple campus LAN used all lecture:

```
Student laptop --Wi-Fi--> Access point (AP) --cable--> Switch --> Router / firewall --> Internet routers
                                                          |                                    |
                                                  Shared printer                        LMS server
```

Real routes can include more devices.

LAN, WAN, and the Internet

Term	Meaning	Everyday example
LAN (Local Area Network)	A network in a limited local area	Computers and a printer in one lab
WAN (Wide Area Network)	Links networks over larger distances	A university links two distant sites
Internet	Many connected networks worldwide	Your campus reaches a cloud LMS

A campus can contain **many separate LANs**.

Endpoints send and receive data

An **endpoint** is a device at an end of a communication.

- **Client:** starts a request, such as a student's browser.

- **Server:** provides a service, such as course files.

Printers and cameras are also connected devices, so they are also part of the attack surface.

The devices in between

Device	What it does	Security concern
Switch	Forwards **Ethernet frames inside a LAN**. It learns local device addresses called **MAC addresses**.	An **unused cable socket** can be an entry point.
Access point (AP)	Lets **wireless** devices join a network.	**Who can connect, and which network do they reach?**
Router	Forwards **IP packets between networks**. Your **default gateway** is the next router for other networks.	Settings can be changed by an attacker.

A home "Wi-Fi router" often combines **router + AP + switch** in one box, which is why people confuse the three.

> **Switches connect local devices. Routers connect networks.**

Where can a network risk begin?

Location	What could go wrong?	Useful control
Device	Malware uses the device's connections	Updates and endpoint protection
Wi-Fi or cable socket	An unauthorized device joins	Network access control
Router settings	An attacker changes settings	Protected admin access and updates

A **control** is a measure that reduces a security risk (same meaning as in Week 2).

Real example: WannaCry in 2017

- WannaCry spread to **reachable, unpatched** Windows systems.
- **Where was the weakness?** A vulnerable **SMB** service. SMB supports **file sharing**.
- **What could reduce spread? Install the fix**, and **restrict unnecessary SMB traffic**.

This is the Week 2 case seen from the network side: the same ransomware, but now the question is *how it reached the machines*. **Reachability is part of the risk.**

Quick check

> The Internet connection fails. **Can a local printer still work?**

Yes. Printing is a **local** path (laptop → AP → switch → printer) and never leaves the LAN. Only paths that cross the router to other networks fail. This is why "Wi-Fi" and "the Internet" are different things.

Activity: label the network

> Name A–D, then trace a print job and an LMS request. Add one risk beside a device or connection and explain why it belongs there.

Answers: A = **access point**, B = **switch**, C = **Internet routers**, D = **router / firewall**.

- **Print job:** laptop → AP → switch → printer. **Printing can stay local.**
- **LMS request:** laptop → AP → switch → router/firewall → Internet routers → LMS server. **The remote LMS path crosses other networks.**
- **Example risk:** an unknown device joins Wi-Fi. **Example control:** require approved network access.

2. Read a network connection

> An **address** identifies the destination. A **service** follows communication rules.

An IP address helps deliver packets

An **Internet Protocol (IP) address** is a number for a **device's network connection (interface)**.

Example item	Address in the classroom model
Student laptop	192.168.10.24
Default gateway	192.168.10.1
Remote LMS server	198.51.100.20

> **An IP address does not prove a person's identity.** (Addresses are shared, reassigned and translated.)

Private addresses and NAT

Inside the LAN, laptops have **private** addresses (192.168.10.24, 192.168.10.25). The router has a **public** side (203.0.113.5).

NAT = Network Address Translation. It **changes address information as traffic passes**, so many private devices share one public address. A result worth remembering: from the outside, those devices look like one address.

A port identifies a service

One IP address can offer **several services**, so the port says which one:

Port	Service
443	HTTPS website
22	SSH administration

> **IP address: which network destination? Port number: which service at that destination?**

Protocols define communication rules

Protocol	Plain English	In the web example
IP	Delivers packets between networks	Destination address
TCP	Delivers data **reliably and in order**	Common transport for HTTP/1.1 and HTTP/2
TLS (Transport Layer Security)	Protects a connection's data	Encryption and certificate checks
HTTP (Hypertext Transfer Protocol)	Defines web requests and responses	Request a course page

- **HTTPS is HTTP protected by TLS.**
- **Certificate:** digital evidence used to **check the server's identity**.

DNS looks up a name

DNS = Domain Name System. A **resolver** finds and returns DNS information.

1. Device asks: **lms.example.com?**
2. Resolver answers: **198.51.100.20**

3. Device connects to **198.51.100.20**

The URL describes a web request

`https://lms.example.com/course/6`

Part	Meaning
`https`	Use HTTP protected by TLS
`lms.example.com`	**Domain name.** DNS can find its address
`/course/6`	Path to a resource on the site
(port 443)	Default HTTPS port when no port is written

> **Check the full domain name before signing in.** (Week 3's hostname lesson, now explained by DNS.)

HTTP and HTTPS on the network

HTTP	HTTPS
No TLS protection for the web content	TLS protects the web content **in transit**
Someone on the path **may read or change it**	Someone on the path **cannot normally read or change it**
Readable content, e.g. `course=6&student=demo`	A network observer sees **encrypted** content when TLS works correctly

Encryption makes content unreadable without the right keys.

HTTPS has a clear protection boundary

Question	What HTTPS tells us
Can a nearby observer read the page content?	**Normally no**, when TLS works correctly
Is every website using HTTPS honest?	**No.** A phishing site can also use HTTPS
Can malware read data on my own device?	**Yes.** Endpoint security still matters
Is all network information hidden?	**No.** IP addresses and traffic timing may remain visible

This table is the heart of the lecture: HTTPS protects **content in transit between the TLS endpoints**, and nothing else.

Demo: a page makes network requests

Open DevTools → **Network** → reload the page. **A row is a resource request, not one network packet.** Clicking a request and choosing **Headers** shows the **Request URL** (the requested address) and **GET** (a request for content).

The browser can show clear content **because it is one endpoint** — decryption happens there.

Quick check

> Which term answers each question?

Question	Term
Where is it?	**IP address**
Which service?	**Port**
Which rules?	**Protocol**
What address matches the name?	**DNS**

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3. Locate and reduce risk

> A useful explanation names the **location**, the **weakness**, and the **control**.

Security matters along the whole path

Five locations, each with its typical control:

#	Location	Control that applies there
1	**Device**	Antivirus / endpoint protection, updates
2	**Access** (Wi-Fi, sockets)	Authenticated Wi-Fi, network access control
3	**Boundary** (router/firewall)	Firewall rules
4	**Data in transit**	TLS (HTTPS)
5	**Server**	Permissions, authentication and authorization

> **Point to the location before naming the control.**

Risk at access: a fake hotspot

Café scenario: two networks advertise the **same familiar name**.

Wi-Fi name (SSID)	What the name proves
Campus_Guest	Only the advertised name
Campus_Guest	It could belong to another operator

Evil twin: a fake access point that **imitates a trusted Wi-Fi network**.

Response: verify the network, follow approved setup instructions, and **stop at certificate warnings** — which answers the opening café question.

Risk in DNS: a false destination

DNS spoofing supplies **false DNS information**.

Question	Expected answer	False answer
lms.example.com?	198.51.100.20 (intended service)	198.51.100.50 (wrong service)

A false DNS answer **can redirect traffic**, but **it does not automatically defeat TLS**: the fake server still has to present a certificate for that name, which is why the warning appears.

Risk in transit: an on-path attacker

On-path attacker: someone positioned to **intercept or influence traffic** (browser → network intermediary → web server).

- **HTTP content can be exposed.**
- **Correct HTTPS protects content between its TLS endpoints.**

Risk at a service: too much exposure

> A server should expose **only the access it needs**.

Connection	Example rule	Reason
Public users to **HTTPS 443**	**Allow** required website traffic	Students need the LMS
Public users to **SSH 22**	**Deny** public administration	Limit admin access
Approved admin path to **SSH**	**Allow** authorized admins	Maintain the server securely

Firewall: software or hardware that **allows or blocks traffic using rules**.

Risk at the server: excessive access

The LMS receives an **encrypted** request. It must still decide **what the user may read**:

- **Authentication:** *Who are you?* The system checks your **identity**.
- **Authorization:** *What may you do?* The system checks your **permissions**.

Encryption in transit says nothing about permissions at the end. (Compare Week 3: **recognition is not authorization**.)

Segmentation limits unnecessary paths

Segmentation divides a network into groups and controls communication between them.

- Guest devices → staff records server: **DENY**
- Guest devices → Internet access: **ALLOW**

A risk explanation needs a location

Location	Weakness or concern	Control that fits
Laptop	Unpatched software	Install supported updates
Wi-Fi access	Unknown network operator	Verify the approved connection
Web traffic	Sensitive data over HTTP	Use valid HTTPS
Private service	Access from unnecessary users	Limit reachability and permissions

Activity: a small office network

> A small office shares one network. (1) Guests connect to Wi-Fi for Internet access. (2) Staff use a payroll server; a printer runs old firmware. (3) The server allows administration from any Internet address. Draw the path, mark two risks, place two controls, and explain how each helps.

One defensible answer:

Risk	Control	Why it helps
Guests can reach staff systems	**Separate guest access** and enforce traffic rules	Limits access to private services
Administration is public	Allow only an **approved admin path**	Reduces unnecessary exposure
Printer firmware is old	**Update it**, or restrict it while arranging replacement	Reduces exploitable weaknesses

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Q&A: check your understanding

1. What is the difference between a LAN and a WAN? A LAN covers a local area. A WAN links networks over larger distances.

2. What does an access point do? An AP lets wireless devices join a network.

3. How do a switch and a router differ? A basic switch forwards **within** a LAN. A router forwards **between** networks.

4. What does an IP address identify? A network interface, for packet delivery.

5. What does a port number identify? A logical port identifies an **application or service endpoint**.

6. What does DNS do when you open a website? It can return an **address** for the website's **domain name**.

7. What protection does HTTPS add? TLS protects web content **in transit** and **checks the server certificate**.

8. Can a phishing website use HTTPS? Yes. HTTPS alone does not prove the site's owner is honest.

9. Why does an exposed service need attention? A reachable service may allow access or expose a weakness. Limit and maintain it.

10. How can segmentation reduce harm? It limits unnecessary communication and can **restrict an incident's spread**.

Assignment: your own network path

> Choose one: opening a course page, making a video call, or using a shared printer.
>
> 1. Draw the endpoints and the network devices in between.
> 2. Explain where IP addresses, ports, or DNS help in your example.
> 3. Mark **two** possible risks and add a suitable control for each. Submit one PDF to LMS: **diagram + short explanations**.

How to approach it:

- Keep the diagram in the lecture's order: **device → AP → switch → router/firewall → Internet → server**. A printer job stops at the switch; that is a good contrast to show you understand local vs remote.
- For part 2, say which question each term answers: **IP = where, port = which service, DNS = which address matches the name**.
- For part 3, use the **location → weakness → control** pattern, and pick risks at **different** locations (e.g. evil-twin Wi-Fi at access, public SSH at the service) so the two controls aren't the same idea twice.

Sources from the slides: Cisco (switch vs router; network segmentation), Cloudflare (What is DNS?; What is HTTPS?), Microsoft (WannaCrypt, 12 May 2017), Google (Inspect network activity).

Most important for exams

- **Five ideas the lecture ends on:** Wi-Fi connects you to a **network**, which is not the same as the Internet; **switches connect local devices, routers connect networks**; **IP, ports and protocols answer different questions**; **DNS finds an address, HTTPS protects content in transit**; controls are needed **at the device, on the path, and at the service**.
- **LAN / WAN / Internet** definitions with an example each.
- **Switch** = frames inside a LAN, learns **MAC addresses**. **AP** = wireless devices join. **Router** = IP packets between networks; **default gateway** = next router. A home box combines all three.
- **IP = which destination, port = which service** (443 HTTPS, 22 SSH), **protocol = which rules** (IP, TCP, TLS, HTTP). **HTTPS = HTTP + TLS. An IP address does not prove identity.**
- **NAT** translates private addresses to a public one.
- **DNS** maps a **name** to an **address**; **DNS spoofing** gives a false answer but **does not by itself defeat TLS**.
- **HTTPS boundary table** (very likely exam material): protects content in transit; does **not** make a site honest, does **not** stop malware on your device, does **not** hide IP addresses and timing.
- **Evil twin** = fake AP with a familiar SSID. **On-path attacker** = positioned to intercept or influence traffic.
- **Firewall** = allows/blocks traffic by rules; expose only what is needed. **Authentication (who are you)** vs **authorization (what may you do)**.
- **Segmentation** limits unnecessary paths and an incident's spread.
- **Explain any risk as location → weakness → control.** Expect to label the campus diagram (A = AP, B = switch, C = Internet routers, D = router/firewall).